SPARK

Situated nuclear physics learning through augmented reality and knowledge co-construction

Apprentissage situé de la physique nucléaire grâce à la réalité augmentée et la co-construction des connaissances

Doctoral supervision:
Stéphanie Fleck (Prof., co-advisor, PERSEUS), Frédéric Brochard (DR CNRS, co-advisor, IJL)
Institution that will employ the doctoral student: Université de Lorraine
Thesis funded: YES **- Associated project:** i-SITE Impact DeNAMISE
Field: Interdisciplinary (Social Sciences/Physics/Computer Science)
Host laboratory: PERSEUS – interdisciplinary research unit (ergonomics, social psychology, learning science, computer science) within the SHS-CLCS cluster at the University of Lorraine.
Travel within the research units associated with the project is to be expected

Abstract

As part of the University of Lorraine's i-SITE Impact DeNAMISE project, this interdisciplinary thesis project, SPARK (for Situated Physics learning through Augmented Reality and Knowledge co-construction), aims to design and evaluate new teaching tools for nuclear physics education. Learning these concepts remains difficult due to their high level of abstraction, the invisibility of the phenomena, and the constraints associated with experimenting with radioactive sources. The project proposes using mixed reality technologies to design hybrid practical work that allows students to visualize and manipulate nuclear processes that are normally inaccessible. Based on theories of experiential and situated learning, as well as a design-based research approach, the research will combine technological and educational design with empirical evaluation. A prototype hybrid environment will be developed to simulate fundamental nuclear physics experiments and support collaborative learning and scientific inquiry. Experiments will then be conducted to evaluate its educational potential and its impact on student learning and engagement. The expected results include developing design principles for immersive learning environments, improving nuclear physics instruction, and contributing to scientific discourse on issues related to nuclear energy.

Keywords: situated cognition, spatial computing for learning, inquiry-based learning, research in nuclear physics education, hybrid learning environments, human-computer interaction (HCI), Design-based research

Résumé

Dans le cadre du projet i-SITE Impact DeNAMISE de l'Université de Lorraine, ce projet de thèse interdisciplinaire, dénommé SPARK (pour Situated Physics learning through Augmented Reality and Knowledge co-construction), vise à concevoir et à évaluer de nouveaux dispositifs pédagogiques pour l'enseignement de la physique nucléaire. L'apprentissage de ces concepts reste difficile en raison de leur forte abstraction, de l'invisibilité des phénomènes et des contraintes liées à l'expérimentation avec des sources radioactives. Le projet propose de mobiliser les technologies de réalité mixte afin de concevoir des travaux pratiques hybrides permettant de visualiser et de manipuler des processus nucléaires normalement inaccessibles. S'appuyant sur les théories de l'apprentissage expérientiel et situé, ainsi que sur une démarche de *design-based research,* la recherche combinera la conception technologique, le design pédagogique et l'évaluation empirique. Un prototype d'environnement immersif sera développé pour simuler des expériences fondatrices de la physique nucléaire et soutenir l'apprentissage collaboratif et l'investigation scientifique. Des expérimentations seront ensuite menées afin d'évaluer son potentiel pédagogique ainsi que son impact sur les apprentissages et l'engagement des étudiants. Les résultats attendus concernent le développement de principes de conception pour les environnements immersifs d'apprentissage, l'amélioration de l'enseignement de la physique nucléaire et la contribution à la culture scientifique sur les enjeux liés à l'énergie nucléaire.

Mots-clés : cognition située, informatique spatiale pour l'apprentissage, recherche en enseignement de la physique nucléaire, apprentissage par investigation, interaction humain-machine (IHM), recherche par le design

Submission of Applications

Applications, including a CV and cover letter (in French or English), must be sent to the following email addresses: stephanie.fleck@univ-lorraine.fr and frederic.brochard@univ-lorraine.fr, with a copy to olga.chernoburova@univ-lorraine.fr, by July 3, 2026.

Candidate Profile

A student holding a master's degree (or equivalent) in computer science, human-computer interaction, cognitive science, educational technology, physics, or a related field, with a strong interest in science education and immersive learning environments.
The candidate must demonstrate skills or experience in human-computer interaction, virtual/augmented/mixed reality (XR), spatial computing, interactive visualization, or the cognitive sciences of learning. An aptitude for interdisciplinary research, the design of immersive environments, and the empirical evaluation of educational applications is expected.

Minimum language proficiency: C1 in French and B2 or higher in English
Frequent travel between Metz and Nancy is to be expected.

Thesis working conditions

Work location: Research Unit 7312 PERSEUS – Saulcy Campus, Metz
PERSEUS is an interdisciplinary research unit (psychology, ergonomics, education and training sciences, computer science, mechanical engineering). The unit's members are united by their interest in interactions across three main areas: human-computer interactions, human and social interactions, and technology-mediated human interactions, as well as the development of methods for designing and evaluating human experiences arising from these interactions.

Located on the Saulcy campus of the Université de Lorraine, PERSEUS has a participatory design platform that includes an immersive virtual reality space (Hyve3D), a multi-room activity analysis platform featuring biometric measurements, and a dynamic and positive research ecosystem within the Rabelais research space.

Supervising Team

The thesis is part of the Université de Lorraine's DeNAMISE (Development of Nuclear Energy through Advanced Multidisciplinary Innovation and Safety Engineering) project. The doctoral student will benefit from the support of a supervisory team composed of experts in nuclear physics and specialists in instructional design, cognitive ergonomics, and mixed reality affiliated with three laboratories at the Université de Lorraine: PERSEUS, IJL, and LEMTA.

- Stéphanie Fleck, a professor of education and training sciences at the PERSEUS laboratory at the University of Lorraine. she is a recognized specialist in human-computer interaction design and the integration of augmented learning environments in an educational context. Her work on mixed reality and tangible interfaces, particularly within projects such as HOBIT, e-TAC, and Grasp-IT, will provide valuable expertise for the design and evaluation of the innovative educational tools envisaged in this project.

- Frédéric Brochard is a research director in the Fusion Plasmas team at the Jean Lamour Institute and a plasma physicist. He is an experimentalist specializing in controlled thermonuclear fusion and heads a major research platform, SPEKTRE. He will contribute his expertise in nuclear physics and the modeling of physical phenomena to ensure the scientific rigor of the simulations and models integrated into mixed reality environments.

- Nicolas Rimbert, a professor at LEMTA (Laboratory of Energy and Theoretical and Applied Mechanics) at the University of Lorraine, will contribute his expertise in instrumentation and the development of innovative experimental devices. His experience in designing hybrid systems and analyzing heat and radiation transfer will be a major asset for the technical implementation of virtual hands-on experiments.

Description of the PhD Proposal

This interdisciplinary thesis project is part of the Université de Lorraine's i-SITE Impact DeNAMISE project. The project aims to explore alternative forms of educational mediation to make the teaching of nuclear physics a lever for developing transversal competencies (also called 21st-century skills), by explicitly linking it to the energy, environmental, and societal transition challenges specific to this field.

1. Context

The International Atomic Energy Agency (IAEA, 2022) emphasizes the importance of education and training in the nuclear industry for achieving a high level of competence and performance. However, there is a lack of up-to-date, reliable educational resources that can make recent advances in nuclear energy understandable and engaging. Beyond industrial needs, according to UNESCO and the OECD (Ananiadoui & Claro, 2009; Soo Boon, 2019; Tytler & Self, 2020) , supporting scientific literacy in conjunction with the development of transversal competencies (critical and ethical thinking, global and systems thinking, creativity and innovation, information processing and argumentation, collaboration, and problem-solving) is essential to empower every individual to understand the transformations taking place in the world and to participate in decisions regarding human activities, including energy issues. Indeed, issues surrounding nuclear energy carry significant sociopolitical and emotional weight (waste, risks, proliferation, energy mix) and require strong scientific literacy to understand and participate in decisions related to necessary transitions. Unfortunately, learners' misconceptions and persistent fears regarding nuclear topics (Cooper et al., 2003; Morales López & Tuzón Marco, 2022; Sonnberger et al., 2021; Trischler & Bud, 2018) resist traditional teaching methods (Kılınç et al., 2013; Morales López & Tuzón Marco, 2022; Skamp et al., 2019) , in a context of inadequate teacher training programs (Appiah-Twumasi, 2024) and a lack of innovative teaching and learning materials. According to Mor (2025), physics education must undergo profound changes to remain relevant and impactful. For Mor, another central issue that physics departments must address globally is a growing and critical mismatch between academic preparation and the multiple demands of industry, research, and public engagement.

Learning objects in formal and non-formal[1] educational and training contexts (e.g., elementary school, high school, university, vocational training center, museums, outreach centers, etc.) are both conceptual AND experiential. They encompass knowledge, skills, and cultures specific to fields of human activity, reflecting the legacy of diverse culturally situated experiences, the scope and complexity of which continue to grow. The (re)construction of these objects is necessary for learners to know, do, be, and live together in a constantly evolving world. This cultural transfer, therefore, requires the design and implementation of dedicated teaching-learning systems.

2. Innovating for Experiential and Contextualized Teaching and Learning in Nuclear Physics

For decades, physics curricula and the educational systems that deliver them have remained largely static, prioritizing transmission-demonstration teaching methods and leaving little room for active learning.
According to Bruner, cited by Kirschner et al. (2022), "any idea, problem, or body of knowledge can be presented in a form simple enough for any learner to understand it in a recognizable form," but, in the context of nuclear physics, this proves difficult and incomplete.

Focusing teaching and learning on *experiential* approaches seems to be a promising avenue here. The concept of experiential learning was theorized in the 1970s by the American pedagogue and educator David A. Kolb (Kolb & Kolb, 2009; Kolb, 1984). It is based on four principles: concrete experience, reflective observation of that experience, conceptualization, and active re-engagement through experience.
Its transformative potential relies on interactions with the physical and social environment, the learner's reflexivity, and their past experiences to influence future experiences.

In this vein, inquiry-based and hands-on approaches have been shown to significantly reduce misconceptions and support education in fields such as radiation, radiation protection, and energy resources (Otsuji et al., 2014; Pizzolato et al., 2014; van Asseldonk, 2019). Physics laboratory work promotes enjoyment, learning, and motivation among students who are not physics majors (Lee & Sulaiman, 2018; Naval, 2020; Wardoyo & Sunismi, 2024) . This is

[1] For more details on the distinctions and nuances between formal, non-formal, and informal learning, please visit (in French): https://oce.uqam.ca/apprentissage-formel-informel-non-formel-des-notions-difficiles-a-utiliser-pourquoi/

particularly true when learners perceive the value and usefulness of the activity, followed by the effort invested in the experiment (Kácovský et al., 2023).
Nevertheless, in the specific case of nuclear physics, even when cognitive load is supported (Appiah-Twumasi, 2024; Azzaroiha et al., 2025), the focus on theoretical and formal aspects limits learners' ability to contextualize them within a broader socio-technical framework (Mor, 2025) . According to Mor, "this prevents students from perceiving the direct relevance of their knowledge to industrial progress and urgent societal challenges, which can undermine their motivation and engagement. It also affects the majority of students who are not aiming for a career as an academic physicist, but who need to acquire scientific knowledge to become active and critical citizens in a democracy." Furthermore, this is not always sufficient to support students' transversal skills and increase their interest in this field (van Asseldonk, 2019) necessary for a situated understanding of the issues. According to , "a (nuclear) physics curriculum for the 21st century should present knowledge about the physical world as the result of human thought, influenced by historical and cultural context, as well as by passion and bias." She calls for forward-looking work so that, by the end of their courses, students feel a heightened sense of wonder and enthusiasm for the physical world, while appreciating the ingenious methods that humanity has used and continues to use to understand and master it.

To achieve this, physics experiential learning must be *situated* and *scaffolded* (Azzaroiha et al., 2025; Grayson, 2020; Mor, 2025). Situated Learning Theory (SLT) explains that learning emerges when individuals actively engage in the sociocultural practices of a community (Lave & Wenger, 1991, 1996). In such a community, new learners reach expert level because they have more opportunities to practice, share knowledge and discoveries, understand recent developments in physics, technological applications, case discussions with peers and specialists, etc. Laboratory work, including collaborative learning, communities of practice, and communities of inquiry, seems particularly relevant to the teaching of nuclear physics, as it grounds abstract concepts in situated, social, and experimental activities.

3. Hybridizing the learning environment to make learning objects tangible and foster communities of inquiry

In nuclear physics, experiential and situated approaches, while promising, are limited in their implementation (Hakim et al., 2022) due to the complexity and risks involved in making real radioactive sources available (cost, regulations, safety), the difficulty in visualizing invisible phenomena (radiation, particle-matter interactions), the lack of immersion in landmark historical experiments (e.g., Thomson's discovery of the electron, Rutherford's experiment). Furthermore, due to the intangible nature of the phenomena (imperceptible and not directly manipulable), nuclear physics presents significant conceptual complexity (scales, probabilities, radioactivity, models) that requires learning *through* abstraction and modeling.

Technologies that bridge the physical, social, and digital worlds (e.g., mixed and extended reality, spatial augmented reality, tangible interaction) offer a unique opportunity to overcome these limitations. Since the 2000s, new families of digital technologies, known as post-WIMP, have been developing, driven by research in Human-Machine Interaction (HMI) through education and learning (HMI). This is a rapidly evolving family of technologies that enables the fusion of real and virtual worlds to create new environments for interacting with information and others, where physical and digital objects coexist and interact in real time (e.g., through a screen, via an augmented reality – AR – or virtual reality – VR –, via 2D or volumetric projection, or through augmented objects).
These interfaces can expand the possibilities for sensorimotor interactions, but also generate enriched cognitive, emotional, and social experiences. Their rapid technological advances mean that human interactions with the physical world are poised for profound change.
The maturity of many of these interaction-based technologies opens new avenues for addressing previously identified needs.

Numerous examples of hybrid solutions are emerging to support learning, particularly in STEM (e.g., Akçayır & Akçayır, 2017; Arymbekov & Turekhanova, 2025; Bacca et al., 2014; Chen et al., 2016; Chen et al., 2019; Chen, 2006; Cheng & Tsai, 2013; Chien et al., 2010; Cuendet et al., 2013; Da Costa et al., 2019; Dupont et al., 2017; Fleck et al., 2015; Fleck & Simon, 2013; Frey et al., 2017; Furio et al., 2017; Ibáñez & Delgado-Kloos, 2018; Martín-Gutiérrez et al., 2015; Munnerley et al., 2014) and have already demonstrated the potential of such approaches.

Nevertheless, few studies focus on nuclear physics, and further work is needed to design hybrid learning environments capable of training students not only to master techniques and concepts but also to engage, in a reflective and responsible manner, with issues related to contemporary nuclear energy.

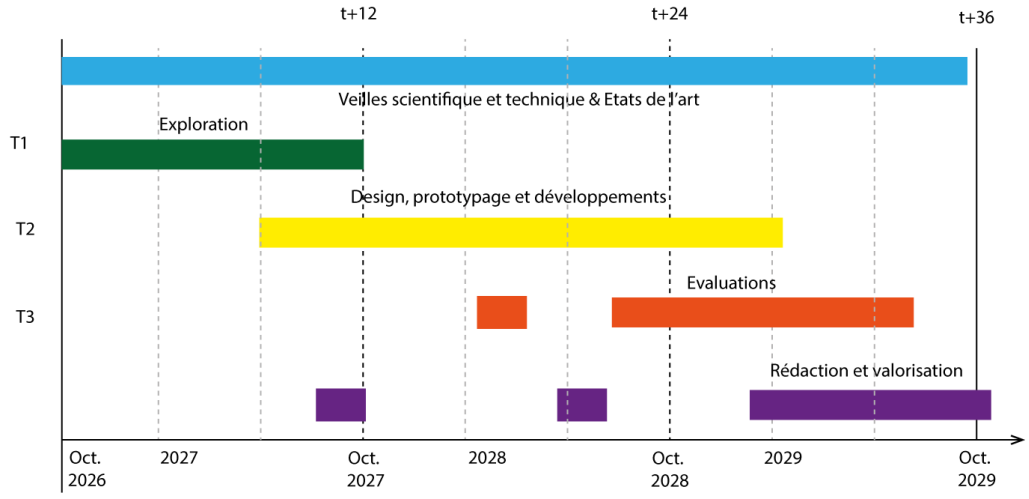
4. Objectives

Within this framework, the SPARK interdisciplinary thesis project (Education Sciences, Physics, Psychology-Ergonomics, and Computer Science) aims to study the necessary dimensions for designing, implementing, and evaluating mixed-reality nuclear physics lab exercises that enable the achievement of three educational objectives:

- Objective 1: To promote the acquisition of key knowledge and skills in nuclear physics through inquiry-based learning centered on fundamental experiments in the field
- Objective 2: To encourage scientific reasoning and the development of critical thinking by fostering the creation of inquiry communities and collaborative learning during lab sessions.
- Objective 3: Place learners in an explicit socio-technical context to **spark** interest, discuss the socio-technical and ethical issues surrounding nuclear energy, and enrich the curriculum beyond mere technical skills

5. Proposed Methodology

This thesis, with an expected duration of three years, is structured around three complementary blocks of research tasks, each contributing to the achievement of the previously defined objectives.
The methodology proposed here is the generative approach of *design-based research* (DBR), particularly used in education and training, as well as in human-machine interaction (Barab & Squire, 2016; Beaudouin-Lafon et al., 2021; Reimann, 2011) .
Thus, while the three research blocks are designed to complement one another, each also constitutes, in and of itself, a source of potential scientific contributions.



T1 - Identifying design spaces through the co-design of a demonstrator usable at various levels of hands-on training
Drawing on the expertise of the project members, this thesis will undertake an exploratory analysis of prior work and existing practices[2] . This work will involve defining physical models, user interfaces, and pedagogical scenarios based on a robust theoretical and empirical foundation.

This stage aims to answer the following specific questions:
- What forms of hybridization should be prioritized to provide tangible, empowering, and eudaimonic epistemic experiences in nuclear physics?
- Which interactional dimensions should be prioritized during their design to foster the establishment of a community of inquiry?

[2] See, for examplehttps://www.radiationliteracy.org/

Mixed methods, combining theoretical and empirical approaches, will be employed here. The analysis of conceptual fields and the compilation of state-of-the-art reviews will support the cross-analysis of results derived from semi-structured interviews and focus groups, cultural probes, and observations conducted with learners and educational experts in physics and computer science (HCI, XRVR) to inform the various aspects of hybridization and interaction design.

T2 – The design of a collaborative mixed-reality hybrid experimentation environment enabling inquiry-based learning experiences based on reference lab exercises, drawing on:
1) enactivist theories of experiential, situated, and distributed learning,
2) the principles of Value-Sensitive Design (VSD) (Cenci & Cawthorne, 2020) and Human-Centered Design (HCD) in HCI for solutions tailored to the capabilities and needs of learners and their teachers, while taking into account significant ethical considerations (safety, simplicity, inclusivity, etc.), and
3) a co-design approach with stakeholders from the DeNAMISE project, specialists in the fields being taught—including nuclear physicists, HCI and mixed reality experts—and, in an innovative way, with students. Co-design will help limit the significant scientific and technical burden and the risk of fragmentation.
The collaborative work will focus on prototyping the experimental setup (e.g., a mixed-reality table and an interaction-tracking system), integrating real-world data (e.g., radiation spectra measured in the laboratory), and conducting technical tests (latency and simulation accuracy). The bench must include real-time feedback and support (e.g., visualization of particle trajectories and "safety" alerts) and enable a collaborative approach (e.g., working in pairs or teams within a shared environment).

T3 – Evaluation and validation of the educational potential of the proposed solutions in context
1) Definition of metrics and tools for validation
2) Implementation of an iterative phase of empirical evaluations, first under laboratory conditions (user tests), then under quasi-controlled conditions (with student volunteers), and finally in real-world usage scenarios. This work will assess i) the device's ergonomic and pedagogical quality, ii) its pedagogical potential, through the evaluation of acquired and experienced learning outcomes, as well as iii) its transformative effects on teaching practices and students' beliefs and interests regarding nuclear energy.

The thesis work will be conducted in accordance with the FAIR principles of research and with a commitment to making the results and outputs openly available.

Originality of the Project

The proposed work is highly interdisciplinary, at the intersection of prospective instructional design, science education, and nuclear physics, and is based on a technological innovation that uses mixed reality for physically realistic simulations. This interdisciplinarity is a major strength because it allows for the simultaneous exploration of:

- the technological design of the system,
- the cognitive mechanisms of learning,
- the scientific rigor of the physics content.

It also fosters multiple scientific contributions, particularly in science education and human-computer interaction.
By following an SVD design approach centered on learners, teachers, and experts, this thesis project falls within the emerging field of Learning Experience Design (LXD). This methodology enables iterative and participatory design, a strong link between theory and practice, and empirical validation in real-world contexts.

The project's originality also lies in its historical grounding through the reinterpretation of foundational experiments using modern, engaging, and safe techno-pedagogical tools:

- Based on precise physical models (e.g., simulation of radioactive decay, radiation-matter interaction).
- Recreating historical experiments (e.g., bubble chamber, Geiger counter, Rutherford experiment).
- Using fictional sources and virtual instruments (e.g., radiation detectors, protective screens).
- Furthermore, unlike many individual simulations, the project aims to foster:
- Collaborative learning,
- Collective knowledge building,
- Scientific reasoning.

- This brings the system closer to a simulated scientific research community, which constitutes an educational innovation.

Applications and expected outcomes

For research:

- Increased knowledge regarding the learning of abstract concepts in physics
- Increased knowledge of HMI design for learning
- Expansion of pedagogical hybridization models
- Contribution to the study of the effectiveness of hybrid environments in science
- Development of new methods for evaluating learning experiences in hybrid contexts

For teaching:

- Development of transferable teaching and learning methods in nuclear physics.
- Wider access to nuclear physics lab sessions (in institutions without dedicated labs, such as museums and science centers).
- Reduction of costs and risks associated with handling radioactive sources.
-

For society:

- Improvement of scientific literacy
- Raising public awareness of issues related to radioactivity (e.g., museums, interactive exhibits).
- Contribution to training for careers in the nuclear sector, which are expected to grow to meet carbon neutrality goals.

Références

Akçayır, M., & Akçayır, G. (2017). Advantages and challenges associated with augmented reality for education: A systematic review of the literature. *Educational Research Review*, *20*, 1-11.

Ananiadoui, K., & Claro, M. (2009). 21st century skills and competences for new millennium learners in OECD countries.

Appiah-Twumasi, E. (2024). Scaffolding as a cognitive load reduction strategy for teaching atomic and nuclear physics. *Momentum: Physics Education Journal*, *8*(2), 194-209.

Arymbekov, B., & Turekhanova, K. (2025). Augmented reality in Physics education: A tool for intellectual learning. *Contemporary Educational Researches Journal*, *15*(1), 33-44.

Azzaroiha, C., Redhana, I. W., & Suma, K. (2025). The Effect of Scaffolding Strategies on Learning Outcomes in Science Learning: A Systematic Literature Review. *Jurnal Penelitian Pendidikan IPA*, *11*(1), 45-55. https://doi.org/10.29303/jppipa.v11i1.8628. https://jppipa.unram.ac.id/index.php/jppipa/article/view/8628(

Bacca, J., Baldiris, S., Fabregat, R., & Graf, S. (2014). Augmented reality trends in education: a systematic review of research and applications. *Journal of Educational Technology and Society,*, *17*(4), 133-149. http://hdl.handle.net/10256/17763 http://hdl.handle.net/10256/17763(

Barab, S. A., & Squire, K. (2016). *Design-based research: Clarifying the terms. A special issue of the journal of the learning sciences*. Psychology Press.

Beaudouin-Lafon, M., Bodker, S., & Mackay, W. E. (2021). Generative Theories of Interaction. *ACM Trans. Comput.-Hum. Interact.*, *28*(6). https://doi.org/10.1145/3468505. https://doi.org/10.1145/3468505(

Cenci, A., & Cawthorne, D. (2020). Refining value sensitive design: A (capability-based) procedural ethics approach to technological design for well-being. *Science and Engineering Ethics*, *26*(5), 2629-2662.

Chen, C.-H., Chou, Y.-Y., & Huang, C.-Y. (2016). An Augmented-Reality-Based Concept Map to Support Mobile Learning for Science [journal article]. *The Asia-Pacific Education Researcher*, *25*(4), 567-578. https://doi.org/10.1007/s40299-016-0284-3. https://doi.org/10.1007/s40299-016-0284-3(

Chen, C.-H., Huang, C.-Y., & Chou, Y.-Y. (2019). Effects of augmented reality-based multidimensional concept maps on students' learning achievement, motivation and acceptance. *Universal Access in the Information Society*, *18*(2), 257-268. https://doi.org/10.1007/s10209-017-0595-z. https://doi.org/10.1007/s10209-017-0595-z(

Chen, Y.-C. (2006, (14-17 juin)). *A study of comparing the use of augmented reality and physical models in the chemistry education* VRCIA '06 Proceedings of the 2006 ACM international conference on Virtual reality continuum and its applications, Hong Kong. pp. 369 - 372,

Cheng, K.-H., & Tsai, C.-C. (2013). Affordances of Augmented Reality in Science Learning: Suggestions for Future Research. *Journal of Science Education and Technology*, *22*(4), 449-462. https://doi.org/10.1007/s10956-012-9405-9. http://dx.doi.org/10.1007/s10956-012-9405-9(

Chien, C.-H., Chen, C.-H., & Jeng, T.-S. (2010). *An interactive augmented reality system for learning anatomy structure*. proceedings of the international multiconference of engineers and computer scientists, 1,

Cooper, S., Yeo, S., & Zadnik, M. (2003). Australian students' views on nuclear issues: Does teaching alter prior beliefs? *Physics Education*, *38*(2), 123-129.

Cuendet, S., Bonnard, Q., Do-Lenh, S., & Dillenbourg, P. (2013). Designing augmented reality for the classroom. *Computers & Education*, *68*(0), 557-569. https://doi.org/http://dx.doi.org/10.1016/j.compedu.2013.02.015. http://www.sciencedirect.com/science/article/pii/S0360131513000547(

Da Costa, J., Szilas, N., & Mueller, A. (2019). *Réalité augmentée pour l'apprentissage conceptuel en sciences: quels principes de conception pour les EIAH? Cas du dispositif DEAPE LEARN en électromagnétisme*. Actes de la 9ème Conférence sur les Environnements Informatiques pour l'Apprentissage Humain, Paris, France.Vol.

Dupont, L., Pallot, M., Morel, L., Christmann, O., Boly, V., & Richir, S. (2017). Exploring Mixed-methods Instruments for Performance Evaluation of Immersive Collaborative Environments. *International Journal of Virtual Reality*, *17*(2), 1-29. https://doi.org/10.20870/IJVR.2017.17.2.2888. https://ijvr.eu/article/view/2888(

Fleck, S., Hachet, M., & Bastien, J. M. C. (2015, June 21-24). *Marker-based augmented reality: Instructional-design to improve children interactions with astronomical concepts*. ACM SIGCHI 14th International Conference on Interaction Design and Children, Boston, MA USA.Vol. pp. 21-28, https://doi.org/10.1145/2771839.2771842

Fleck, S., & Simon, G. (2013). *An Augmented Reality Environment for Astronomy Learning in Elementary Grades: An Exploratory Study* 25ème ACM conférence francophone sur l'Interaction Homme-Machine-IHM 2013, Bordeaux, France. pp. 14-22, https://doi.org/10.1145/2534903.2534907

Frey, J., Gervais, R., Lainé, T., Duluc, M., Germain, H., Fleck, S., Lotte, F., & Hachet, M. (2017). *Scientific outreach with Teegi, a tangible EEG interface to talk about neurotechnologies*. Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems, pp. 405-408,

Furio, D., Fleck, S., Bousquet, B., Guillet, J.-P., Canioni, L., & Hachet, M. (2017). *HOBIT: Hybrid Optical Bench for Innovative Teaching* 2017 CHI Conference on Human Factors in Computing Systems, Denver, CO USA. pp. 949-959, https://doi.org/10.1145/3025453.3025789

Grayson, D. J. (2020). *Physics education for 21st century graduates*. Journal of Physics: Conference Series, 1512, pp. 012043,

Hakim, L., Lubis, P. H. M., Lefudin, L., & Sulistyowati, R. (2022). The relationship between learning style toward mastery of concepts and problem solving skill introduction to nuclear physics for pre-service teacher. *Jurnal Penelitian Pendidikan IPA*, *8*(2), 666-673.

Ibáñez, M.-B., & Delgado-Kloos, C. (2018). Augmented reality for STEM learning: A systematic review. *Computers & Education*, *123*, 109-123. https://doi.org/https://doi.org/10.1016/j.compedu.2018.05.002. http://www.sciencedirect.com/science/article/pii/S0360131518301027(

Kılınç, A., Boyes, E., & Stanisstreet, M. (2013). Exploring Students' Ideas About Risks and Benefits of Nuclear Power Using Risk Perception Theories. *Journal of Science Education and Technology*, *22*(3), 252-266. https://doi.org/10.1007/s10956-012-9390-z. https://doi.org/10.1007/s10956-012-9390-z(

Kolb, A. Y., & Kolb, D. A. (2009). Experiential learning theory: A dynamic, holistic approach to management learning, education and development. *The SAGE handbook of management learning, education and development*, 42-68.

Kolb, D. A. (1984). *Experiential learning.* Englewood Cliffs, New Jersey: Prentice-Hall.

Lave, J., & Wenger, E. (1991). *Situated learning: Legitimate peripheral participation*. Cambridge university press.

Lave, J., & Wenger, E. (1996). Communities of practice. In : Bron: http://www. infed. org/biblio.

Lee, M. C., & Sulaiman, F. (2018). The effectiveness of practical work on students' motivation and understanding towards learning Physics. *International Journal of Humanities and Social Science Invention*, *7*(8), 2319-7714.

Martín-Gutiérrez, J., Fabiani, P., Benesova, W., Meneses, M. D., & Mora, C. E. (2015). Augmented reality to promote collaborative and autonomous learning in higher education. *Computers in Human Behavior*, *51*, 752-761.

Mor, B. (2025). Reimagining Physics Education for the 21st Century: A Socio-Technical Perspective on Curriculum Reform and Industrial Relevance. *Schrödinger: Journal of Physics Education*, *6*(3), 152-160. https://doi.org/10.37251/sjpe.v6i3.2013. https://cahaya-ic.com/index.php/SJPE/article/view/2013(

Morales López, A. I., & Tuzón Marco, P. (2022). Misconceptions, knowledge, and attitudes towards the phenomenon of radioactivity. *Science & Education*, *31*(2), 405-426.

Munnerley, D., Bacon, M., Fitzgerald, R., Wilson, A., Hedberg, J., & Steele, J. (2014). Augmented Reality: Application in Higher Education. *Office for Learning and Teaching (Australia). DOI*, *10*(2.1), 3121.7445.

Naval, D. (2020). Practical work: Its effects on performance and motivation of non-physics major students. *International Journal of Research Studies in Education*, *9*(1), 99-108.

Otsuji, H., Toda, M., Nobeoka, A., & Taylor, P. C. S. (2014). *The developed inquiry-based spiral curriculum to promote the radiation literacy as STS and Sustainability Education*. International Conference of STEM in Education 2014,

Pizzolato, N., Fazio, C., & Battaglia, O. R. (2014). Open inquiry-based learning experiences: a case study in the context of energy exchange by thermal radiation. *European Journal of Physics*, *35*(1), 015024. https://doi.org/10.1088/0143-0807/35/1/015024. https://doi.org/10.1088/0143-0807/35/1/015024(

Reimann, P. (2011). Design-Based Research. In L. Markauskaite, P. Freebody, & J. Irwin (Eds.), *Methodological Choice and Design: Scholarship, Policy and Practice in Social and Educational Research* (pp. 37-50). Springer Netherlands. https://doi.org/10.1007/978-90-481-8933-5_3

https://link.springer.com/content/pdf/10.1007%2F978-90-481-8933-5_3.pdf

Skamp, K., Boyes, E., Stanisstreet, M., Rodriguez, M., Malandrakis, G., Fortner, R., Kilinc, A., Taylor, N., Chhokar, K., & Dua, S. (2019). Renewable and nuclear energy: An international study of students' beliefs about, and willingness to act, in relation to two energy production scenarios. *Research in Science Education*, *49*(2), 295-329.

Sonnberger, M., Ruddat, M., Arnold, A., Scheer, D., Poortinga, W., Böhm, G., Bertoldo, R., Mays, C., Pidgeon, N., & Poumadère, M. (2021). Climate concerned but anti-nuclear: Exploring (dis) approval of nuclear energy in four European countries. *Energy Research & Social Science*, *75*, 102008.

Soo Boon, N. (2019). *Exploring STEM competences for the 21st century* (In-Progress reflection on current and critical issues in curriculum, learning and assessment, Issue. https://unesdoc.unesco.org/notice?id=p::usmarcdef_0000368485

Trischler, H., & Bud, R. (2018). Public technology: nuclear energy in Europe. *History and Technology*, *34*(3-4), 187-212. https://doi.org/10.1080/07341512.2018.1570674. https://doi.org/10.1080/07341512.2018.1570674(

Tytler, R., & Self, J. (2020). *Designing a contemporary STEM curriculum* (In-Progress reflection on current and critical issues in curriculum, teaching, learning and assessment, Issue. https://unesdoc.unesco.org/notice?id=p::usmarcdef_0000374146

van Asseldonk, K. K. (2019). *Enhancing students' intrinsic motivation for radiation physics by inquiry-based learning: A quasi-experimental study of student behaviour*

Wardoyo, A., & Sunismi, S. (2024). Contextual Teaching and Learning Approach Based On Tpack To Increase Students' Interest In Learning and Learning Outcomes. *Mathline : Jurnal Matematika dan Pendidikan Matematika*, *9*(3), 629-644. https://doi.org/10.31943/mathline.v9i3.598. https://mathline.unwir.ac.id/index.php/Mathline/article/view/598(